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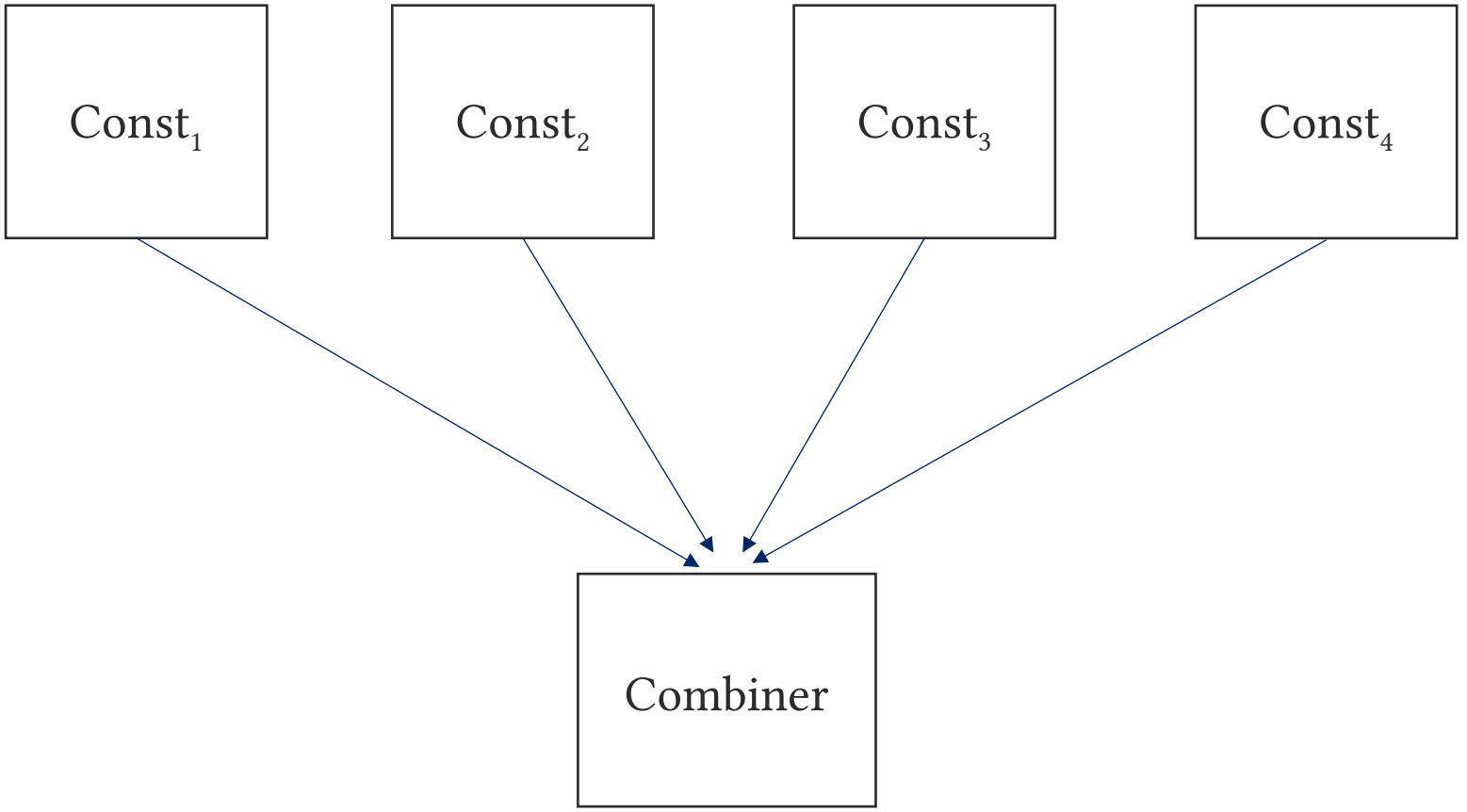
# On Robust Combiners for Pseudorandom Codes and PKE with Pseudorandom Ciphertext

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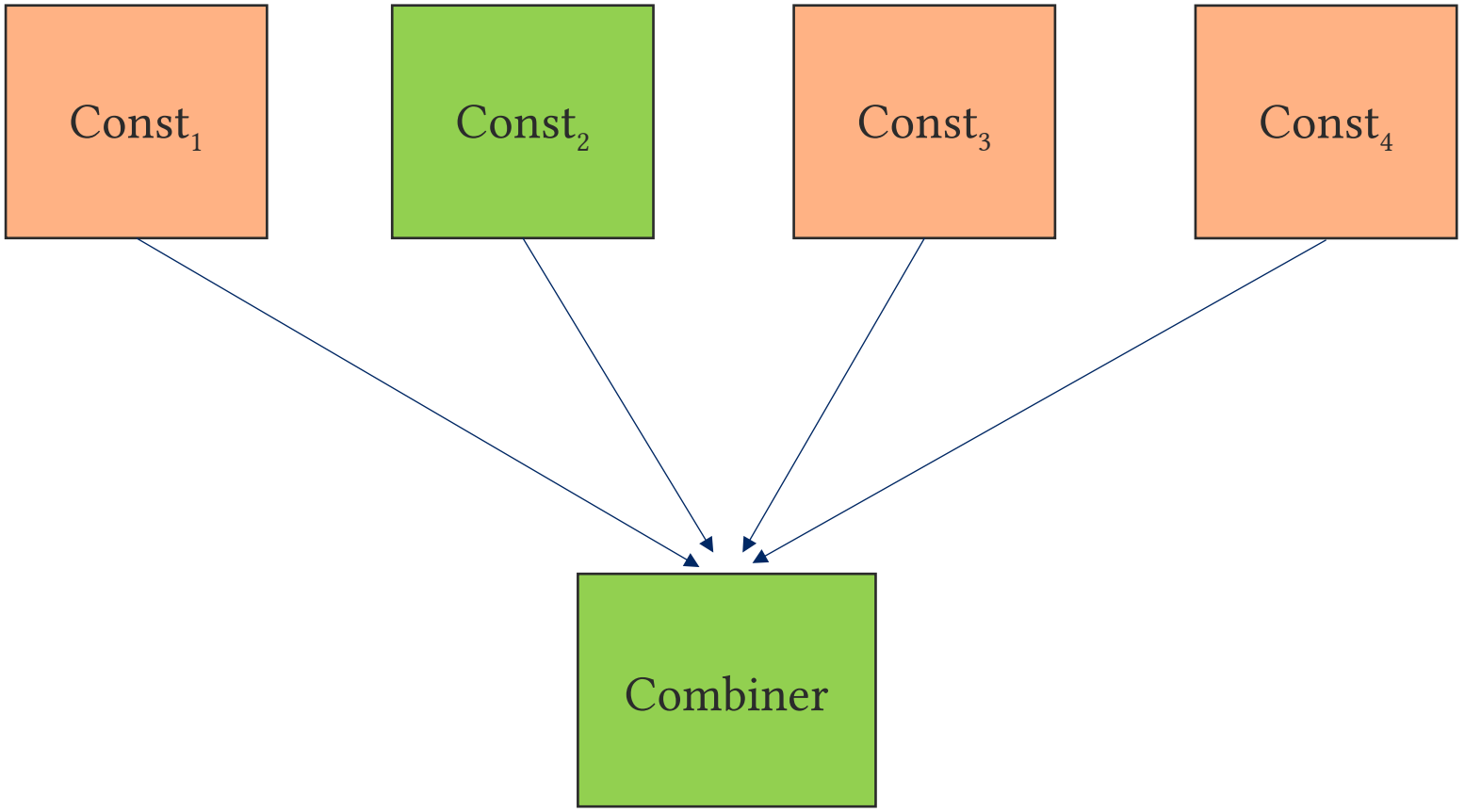
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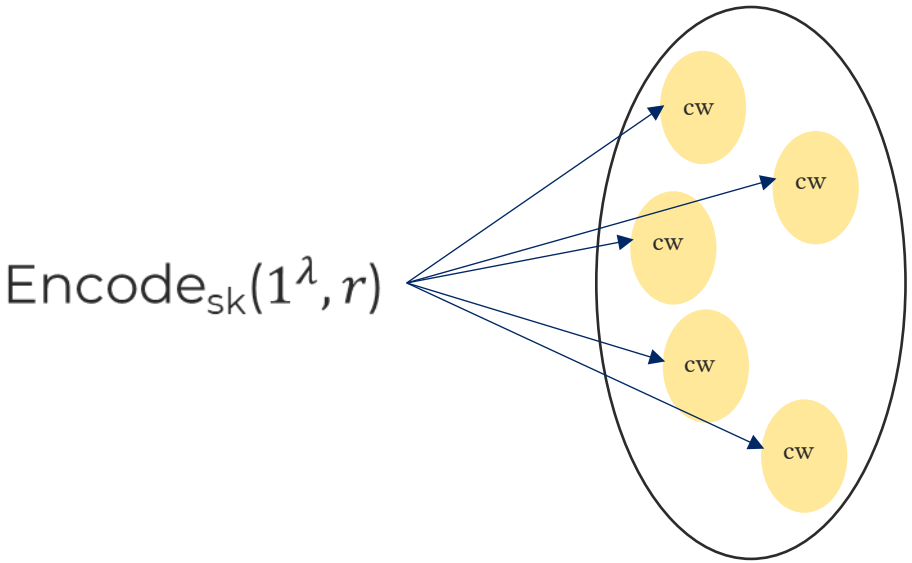
# Robust Combiners



# Robust Combiners



# Pseudorandom Codes




- **Pseudorandomness**

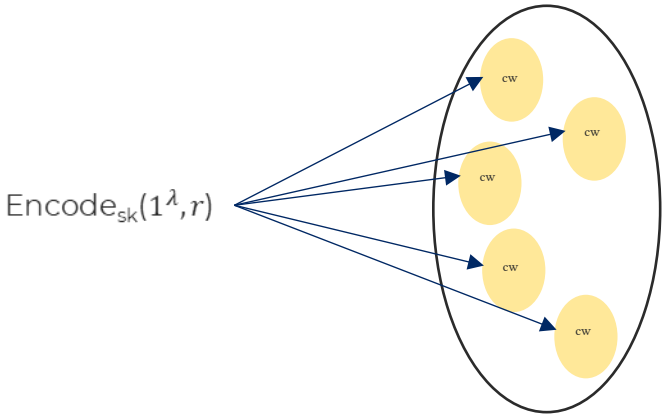
  $\mathsf{cw} \approx U_\lambda$

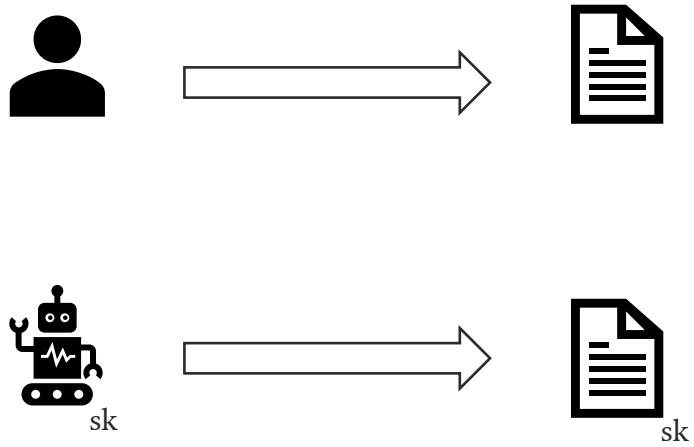
- **Error Correction**

  $\mathsf{Decode}_{\mathsf{sk}}($ $) =$ 

  $\mathsf{Decode}_{\mathsf{sk}}($ $) =$ 

# Pseudorandom Codes for Watermarking




- **Pseudorandomness** ≈
  - $\text{cw} \approx U_\lambda$
- **Error Correction** ≈
  - $\text{Decode}_{sk}(\quad) =$ 👍
  - $\text{Decode}_{sk}(\quad) =$ 👎
- **Quality of**  sk
- **Robust Detection**
  - $\text{Decode}_{sk}($  sk 👍
  - $\text{Decode}_{sk}($  👎

# Constructions of Pseudorandom Codes

## Constructions

- Planted XOR and hyperloop assumptions [GG‘25]
- Local weak pseudorandom functions [GM’24]
- Permutated code assumptions [CGGMW‘25]

## Impossibility Results

- from many generic assumptions [DMR’25, GGW25]

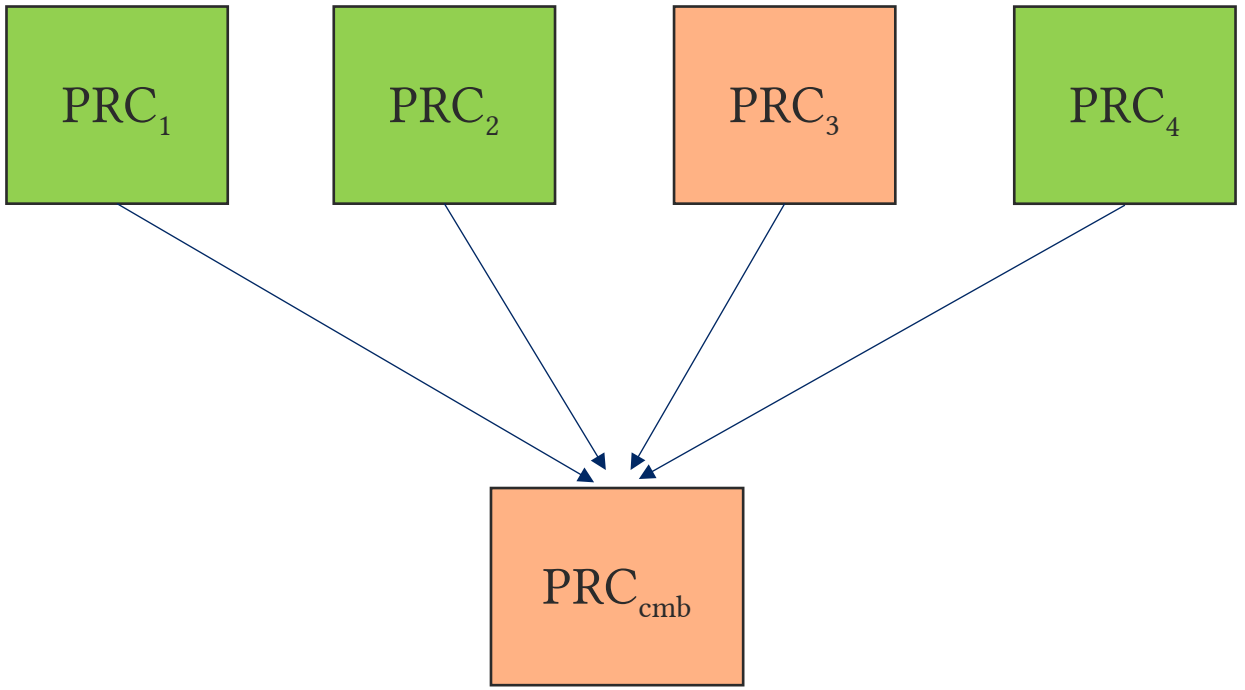


# Can we build a robust combiner for pseudorandom codes?

# Main Results

Main Theorem 1

**Black-box Combiners for secret-key pseudorandom codes do not exist.**



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Main Theorem 1

**Black-box Combiners for secret-key pseudorandom codes do not exist.**

Main Theorem 2

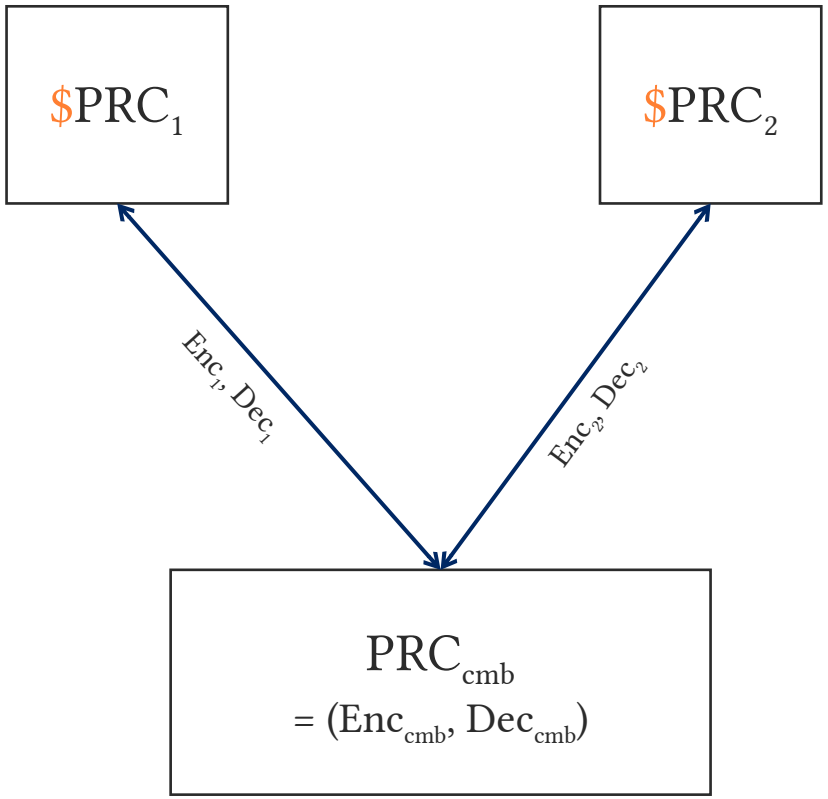
**Black-box combiners for PKE with pseudorandom ciphertexts in the sparse ciphertext regime do not exist.**

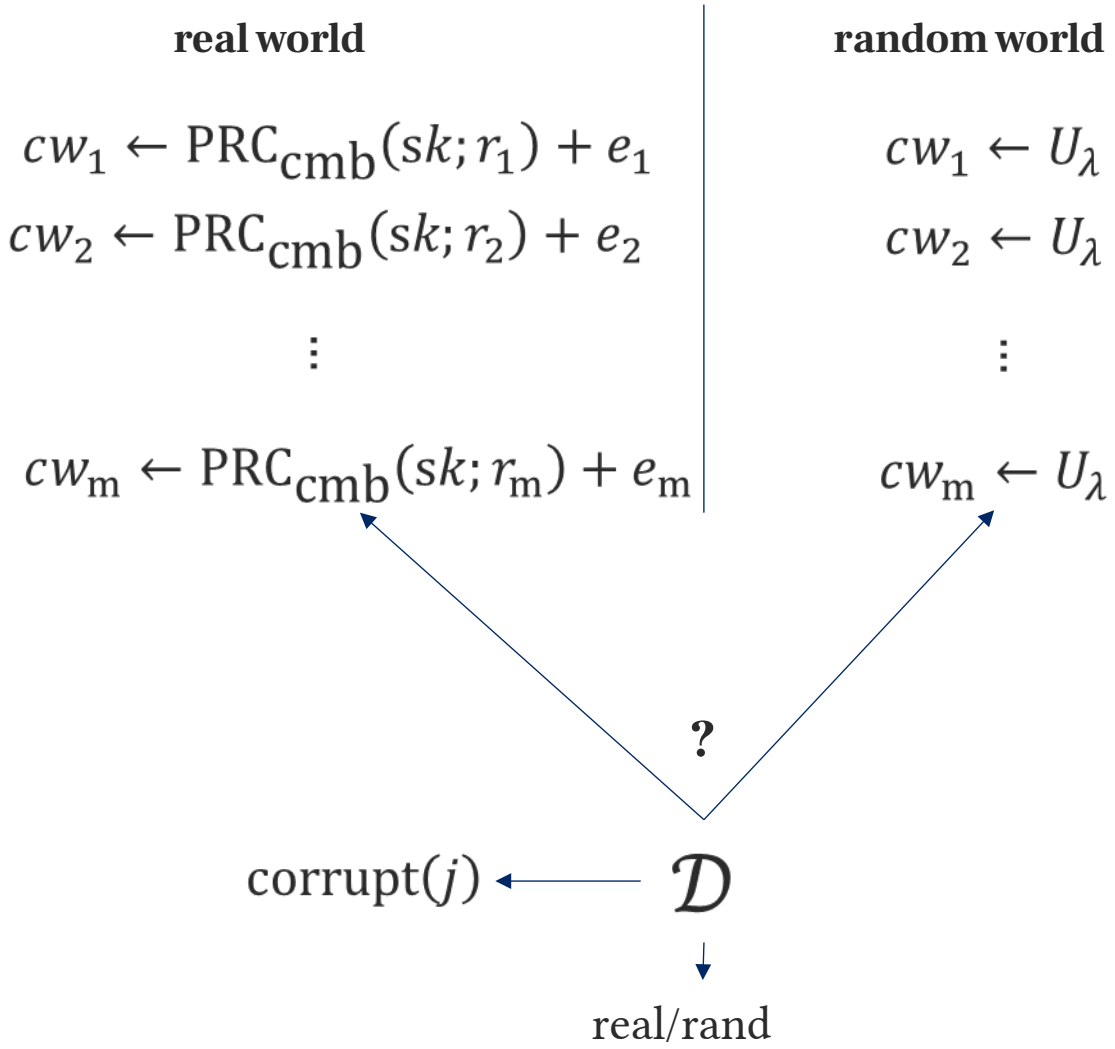


Corollary

**No black-box construction of PKE with pseudorandom ciphertexts in the sparse ciphertext regime from IND-CPA PKE.**

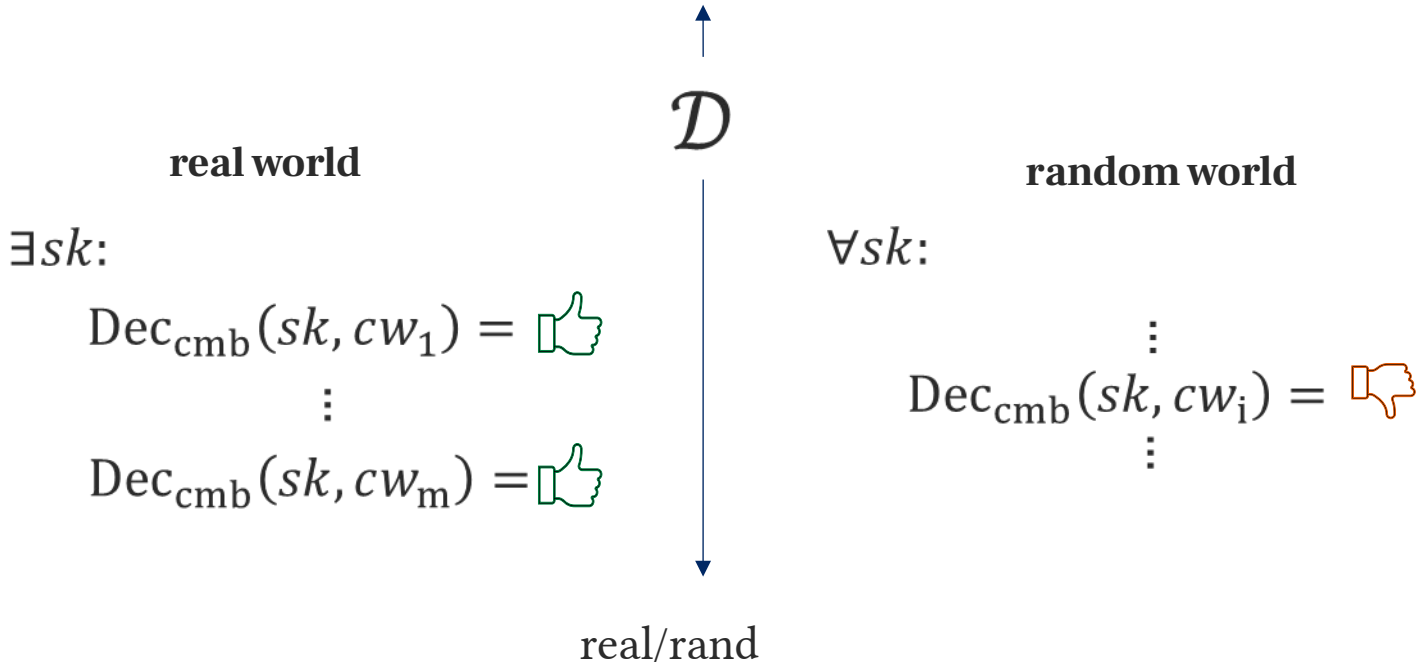
# Modeling Combiners



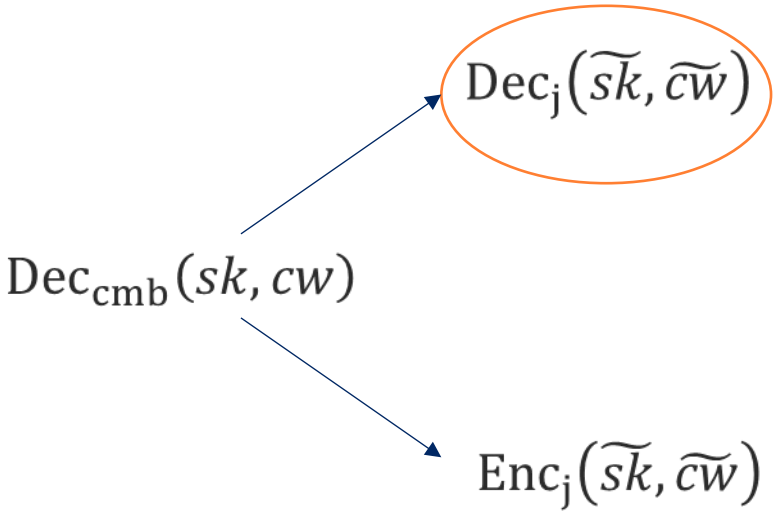



# $\mathrm{Dec}_{\mathrm{cmb}}$ makes no Queries

$$cw_{\mathrm{i}} \leftarrow \mathrm{PRC}_{\mathrm{cmb}}(sk; r_{\mathrm{i}}) + e_{\mathrm{i}} \,||\, U_{\lambda}$$



# First Query of $\mathrm{Dec}_{\mathrm{cmb}}$



# $\mathbf{Dec_{cmb}}$ queries $\mathbf{Dec}_i$

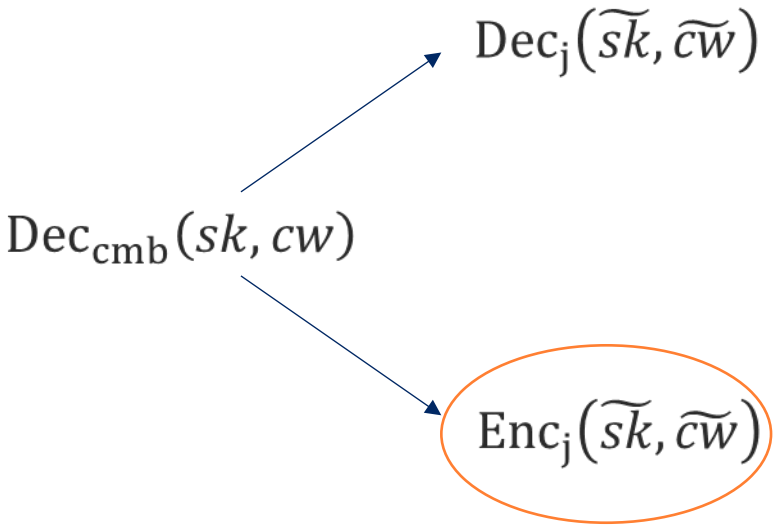
$$\mathrm{Dec}_{\mathrm{cmb}}(sk, cw_i) \longrightarrow \mathrm{Dec}_j(\widetilde{sk}, \widetilde{cw}) = \perp$$

**Claim**: $\mathrm{Dec}_j(\widetilde{sk}, \widetilde{cw}) = \perp$ (w.h.p.)

**real world** **random world**

$$\mathbb{P}\left[\mathrm{Dec}_j(\widetilde{sk}, \widetilde{cw}) \neq \perp\right] \underset{\text{(corrupt } j\text{)}}{\approx} \mathbb{P}\left[\mathrm{Dec}_j(\widetilde{sk}, \widetilde{cw}) \neq \perp\right] \approx \mathrm{negl}(\lambda)$$

# First Query of $\mathrm{Dec}_{\mathrm{cmb}}$



# $\mathbf{Dec_{cmb}}$ queries $\mathbf{Enc_j}$

$\mathrm{Dec}_{\mathrm{cmb}}(sk, cw_i)$ → $\mathrm{Enc}_j(\widetilde{sk}, \tilde{r})$

$\mathrm{Enc}_j(\widetilde{sk}, \tilde{r})$

$cw_i = \mathrm{Enc}_{\mathrm{cmb}}(sk, r) + e_i$

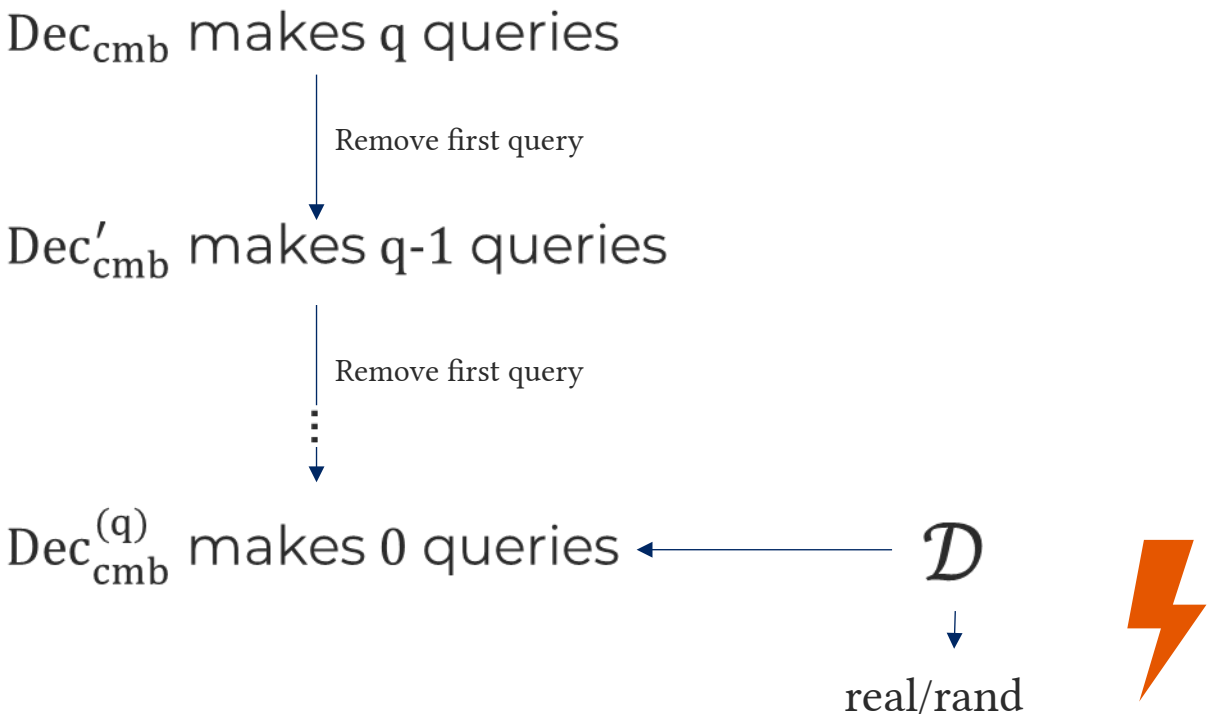
[DMR'25]:

- Learn heavy queries:

$$aux \leftarrow \mathrm{LearnEnc}(sk)$$
$$sk' \leftarrow (sk, aux)$$

- Simulate non-heavy locally

# Distinguisher against Pseudorandomness



# Conclusion

Main Theorem 1

**No Combiners for PRC**

Main Theorem 2

**No Combiners for pseudorandom**
**✦ sparse PKE**

Corollary

**IND-CPA PKE ⇏ pseudorandom**
**+ sparse PKE**

# Thank you!

https://ia.cr/2026/2650